Book strip

Science · GROW · Autumn 2 · Week 1 · GROW_SCI_A2_W01_P1

Day and night: introduce and explore · Weekly arc: Introduce and explore.

Goal: I can explain how Earth's rotation causes day, night, and time zones, and how Earth's tilted axis causes changing day length and seasons.

Today I can: I can identify the lit side of Earth as day and sequence one place through sunrise, middle of the day, sunset, and night. · I can use rotation and apparent movement to explain what an observer on Earth sees across 24 hours. · I can link Earth's 23.5° tilted axis and 15° per hour spin to time zones, length of day through the year, and seasons.

Made by Matt

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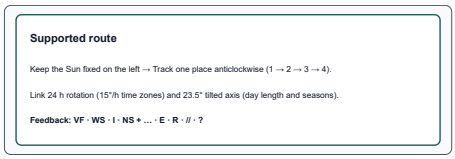
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Route card

Staff: cut these cards apart before the lesson. Give one route card. Mark actual **I · S · C** beside each task.

Supported



Supported route card: track one marked place through sunrise, middle of the day, sunset, and night.

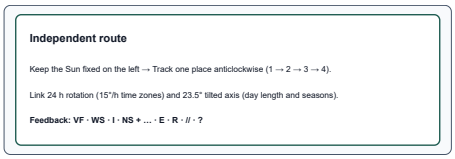
Use Reference A (North-Pole view with the fixed Sun on the left). Order the four daily moments starting at Position 1: Night, Sunset, Sunrise, Middle of the day.

Use Reference A and Reference B. Work in your book or usual mode.

Staff I ☐ S ☐ C ☐ · Feedback: VF ☐ WS ☐ E ☐ NS + … ☐ R ☐ // ☐ ? ☐

My response:

Independent



Independent route card: track one marked place through sunrise, middle of the day, sunset, and night.

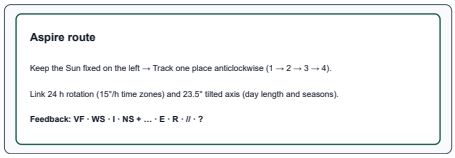
Use Reference A. Track one marked place through Positions 1, 2, 3, and 4 in the North-Pole view. Name the time of day at each position and state the evidence clue.

Use Reference A and Reference B. Work in your book or usual mode.

Staff I ☐ S ☐ C ☐ · Feedback: VF ☐ WS ☐ E ☐ NS + … ☐ R ☐ // ☐ ? ☐

My response:

Aspire



Aspire route card: track one marked place through sunrise, middle of the day, sunset, and night.

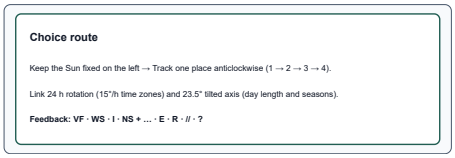
Explain how Earth's west-to-east rotation causes both the apparent east-to-west motion of the Sun and global time zones (using 360° ÷ 24 hours = 15° per hour).

Use Reference A and Reference B. Work in your book or usual mode.

Staff I ☐ S ☐ C ☐ · Feedback: VF ☐ WS ☐ E ☐ NS + … ☐ R ☐ // ☐ ? ☐

My response:

Choice



Choice route card: track one marked place through sunrise, middle of the day, sunset, and night.

1. In our North-Pole model with the Sun on the left, which half of Earth is in daylight: the left half or the right half? 2. What movement causes day and night: Earth's rotation or the Sun switching off?

Use Reference A and Reference B. Work in your book or usual mode.

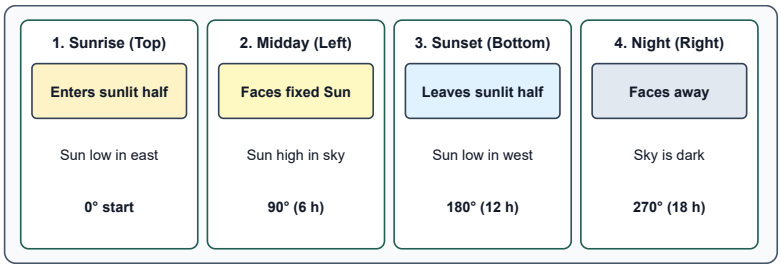
Staff I ☐ S ☐ C ☐ · Feedback: VF ☐ WS ☐ E ☐ NS + … ☐ R ☐ // ☐ ? ☐

My response:

Made by Matt

Reference sheets

Reference A · One marked place rotating through light and dark (North-Pole view)

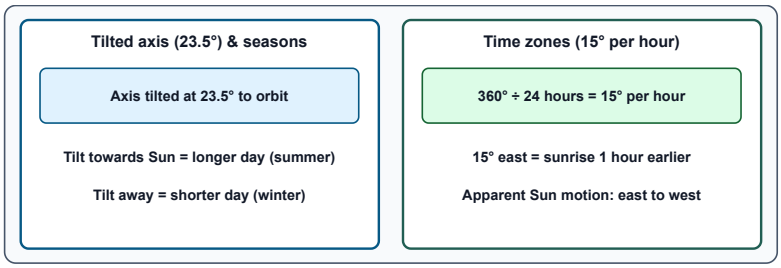


Four positions of a marked place in the North-Pole view with sunlight from the left: 1. Sunrise, 2. Midday, 3. Sunset, 4. Night.

Reference only. Do not write on this sheet. Keep the Sun fixed on the left and follow the red marker anticlockwise (1 → 2 → 3 → 4).

Position in North-Pole view	Time of day	Space viewpoint clue (what the marker does)	Observer viewpoint clue (how the sky looks)
Position 1 · Top boundary	Sunrise	Turns anticlockwise from the unlit half into the sunlit left half	Sun appears low in the eastern sky
Position 2 · Left sunlit side	Middle of the day	Faces directly towards the fixed Sun on the left	Sun appears high across the sky
Position 3 · Bottom boundary	Sunset	Turns anticlockwise out of the sunlit half into the unlit right half	Sun appears low in the western sky
Position 4 · Right unlit side	Night	Faces away from the fixed Sun inside the shaded right half	Sky is dark; the Sun is below the horizon

Reference B · Key Stage 3: tilted axis (23.5°), day length, seasons, and time zones



Key Stage 3 model: Earth's 23.5° tilted axis causes changing day length and seasons, while 360° rotation in 24 hours (15° per hour) causes time zones.

Key Stage 3 concept	Space mechanism (actual cause)	Observable effect on Earth
Apparent Sun motion	Earth rotates west to east on its axis once every 24 hours	The observer turns with Earth, so the Sun appears to move from east to west across the sky
Time zones (15° per hour)	Earth rotates 360° in 24 hours, turning 15° of longitude each hour (360° ÷ 24 = 15°)	Global time zones shift by 1 hour for every 15° of longitude east or west

Key Stage 3 concept	Space mechanism (actual cause)	Observable effect on Earth
Tilted axis & length of day	Earth's axis is tilted at 23.5° relative to its orbit around the Sun	Daylight is longer than 12 hours when our hemisphere tilts towards the Sun and shorter when it tilts away
Seasons	Earth orbits the Sun once a year while keeping its 23.5° axial tilt pointed in the same direction	Hemisphere tilted towards the Sun has summer (longer days, higher Sun); opposite hemisphere has winter (shorter days, lower Sun)

Made by Matt

Catch-up sheet

Absence or access only. This sheet gives a fast entry if a pupil joined late or missed the start. Never add it on top of today's routine tasks.

1. **Fixed Sun, turning Earth:** In our North-Pole model, the Sun stays on the left. The left half of Earth is lit (**day**) and the right half is shaded (**night**).

2. **One rotation (24 hours = 360°):** As Earth rotates anticlockwise on its **axis**, one marked place moves through **Sunrise** (top) → **Middle of the day** (left) → **Sunset** (bottom) → **Night** (right). Turning 15° per hour creates **time zones**.

3. **Tilted axis (23.5°) and seasons:** Earth's 23.5° tilted axis makes daylight longer in **summer** (tilted towards the Sun) and shorter in **winter** (tilted away).

4. **Quick link:** Point to Position 1 and Position 2 on Reference A, say whether the marker is entering light or facing the Sun, then join today's You do Task 1.

Point, speak, sign, draw, type, write, or tell an adult. You may pass.

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Choice route sheet

Circle or point to your choice, then give a short reason from Reference A and Reference B.

1. 1. In our North-Pole model with the Sun on the left, which half of Earth is in daylight: the left half or the right half? 2. What movement causes day and night: Earth's rotation or the Sun switching off? **Staff I ☐ S ☐ C ☐**

Help: Check Reference A and point to or write the matching scientific choice in each pair.

2. 3. When the red marker moves from the lit half into the unlit half at the bottom (Position 3), is that Sunset or Sunrise? 4. How long does one full rotation take: 24 hours or 1 year? **Staff I ☐ S ☐ C ☐**

Help: Use Position 3 on Reference A and the rotation row on Reference B.

3. 5. Why does the Sun seem to cross our sky from east to west: Earth rotates west to east or the Sun orbits Earth daily? Add one short sentence using the word apparent. **Staff I ☐ S ☐ C ☐**

Help: Use Reference B to explain what apparent movement means for an observer on Earth.

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Choice

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt